Uncertainty-Aware Triage for Rule-Based ICAO Face-Image Compliance: Label-Conditional Conformal Guarantees and an Honest Trade-off Study

Mustafa Vedat Yildirim, Inna Skarga-Bandurova
Oxford Brookes University, School of Engineering, Computing and Mathematics
Draft, methodology and results. Introduction and related work positioned for supervisor review.

Abstract

Automated verification of ICAO/ISO compliance for face images in machine-readable travel documents is typically performed by rule-based analyzers that return hard pass/fail decisions per requirement. Such decisions carry no statistical guarantee, and their false rejections translate into unnecessary photo retakes, while a missed violation is far more costly, so the operational problem is triage under class imbalance and asymmetric error cost. We wrap an existing rule-based analyzer (BioGaze) with a lightweight per-criterion supervised meta-classifier followed by conformal calibration, and study, on the TONO synthetic ICAO benchmark under a strictly subject-disjoint protocol, what distribution-free guarantees a multi-criterion compliance pipeline can and cannot deliver. Our study is empirical and diagnostic rather than a new estimator. We report four findings. First, marginal split-conformal holds its overall coverage target (0.90) but silently under-covers the minority violation class on 5 of the 14 criteria (6 if a small-sample criterion is counted), meaning that the whole 95% interval lies below the 0.90 target (as low as 0.40 with an interval of 0.34 to 0.46), and label-conditional (Mondrian) conformal restores per-class coverage to about 0.90 with intervals disjoint from the marginal method on exactly those criteria; intervals are computed on unique test images, so that the overlap between seeds is not counted as independent evidence. Second, through a feature ablation we show the coverage guarantee is feature-agnostic, as theory predicts, whereas most of the apparent false-flag reduction is an artifact of the single-violation dataset structure and is not, by itself, evidence for conformal prediction. Third, across seven triage strategies, including two abstention rules tuned on calibration data only and compared at a matched abstention rate, no method dominates: recalibration supplies most of the accuracy, a conventional abstention rule reaches a lower false-positive rate than Mondrian at the same retake rate, and Mondrian's distinct contribution is the formal class-conditional coverage property: its two thresholds are fixed as calibration quantiles at the nominal level rather than searched against a cost objective. A correctly randomized adaptive prediction set (APS) is competitive rather than degenerate. Fourth, a 14-criterion image-level decision incurs a multiple-comparison problem: naive joint coverage collapses to about 0.27, and a Bonferroni-corrected image-level guarantee of about 0.91 is attainable only by routing almost every image to retake. We confine claims to what TONO can support: it is single-violation, so the ACCEPT class is empty and cannot be validated, which motivates real multi-attribute data as follow-up.

1. Introduction

This paper asks whether the outputs of an existing rule-based ICAO compliance analyzer can be endowed with a statistical guarantee and turned into an uncertainty-aware accept/reject/retake decision, without modifying or retraining the analyzer. The task is not simply to maximise accuracy. In a document-issuance pipeline the cost of accepting a non-compliant photo greatly exceeds the cost of asking for a retake, and true violations are the minority class, so the right framing is uncertainty-aware triage under class imbalance and asymmetric cost, and the right guarantee is one that holds on the violation class, not merely on average.

We use conformal prediction, a distribution-free framework that turns any predictor's scores into prediction sets with a user-chosen coverage level, and we contrast the standard marginal formulation with its label-conditional (Mondrian) variant. Because the same synthetic identities recur across violation types in our data, we adopt a strictly subject-disjoint evaluation to prevent identity leakage between calibration and test. We deliberately treat the contribution as empirical and diagnostic: the marginal-under-covers-then-Mondrian-restores phenomenon is a known property of conformal prediction, and our value is in applying and stress-testing it on a structured, multi-decision compliance pipeline, and in reporting honestly what does and does not transfer.

Contributions.

- A model-agnostic per-criterion supervised meta-classifier with conformal calibration that endows each of 14 ICAO decisions of a rule-based analyzer with a distribution-free coverage guarantee, evaluated under a subject-disjoint protocol with confidence intervals over ten splits.
- Evidence, with disjoint de-duplicated Clopper-Pearson intervals, that marginal conformal under-covers the minority violation class and that label-conditional (Mondrian) conformal restores it, the cost-relevant quantity.
- A feature ablation separating what transfers (the distribution-free guarantee) from what does not (accuracy, which here rides a single-violation dataset artifact), and an honest seven-method trade-off with calibration-only tuning and a matched-abstention comparison, in which recalibration provides most of the accuracy, a conventional abstention rule provides an equal or better empirical operating point, and Mondrian provides the formal class-conditional coverage property.

- An image-level triage analysis that decomposes rejections into correct-target versus wrong-reason, a characterisation of the multiple-comparison problem intrinsic to a 14-criterion decision (a Bonferroni-corrected joint guarantee and its conservatism cost), and an explicit statement of what TONO cannot evaluate.

2. Related work

Conformal prediction provides distribution-free, finite-sample coverage for any base predictor under exchangeability (Vovk, Gammerman and Shafer; Angelopoulos and Bates). We use the least-ambiguous-set (LAC) non-conformity score of Sadinle, Lei and Wasserman, and compare against Adaptive Prediction Sets (Romano, Sesia and Candes). Coverage on a minority class is a known weakness of the marginal formulation; the label-conditional (Mondrian) construction of Vovk and the class-conditional treatments of Ding et al. address it by calibrating within each class. Our per-criterion problem is the binary case of this idea, replicated across 14 coupled criteria, which introduces a joint-coverage (multiple-comparison) question that single-decision studies do not face. The accept/reject/retake structure connects to selective prediction and the reject option (Chow; El-Yaniv and Wiener; Geifman and El-Yaniv), and the asymmetric miss/retake cost connects to conformal risk control (Angelopoulos et al.; Bates et al.), which we use only to reason about an operating point rather than to control a risk functional.

Automated ICAO/ISO compliance verification has been studied primarily as point classification: Ferrara, Franco and Maltoni established conformance checking and the BioLab-ICAO benchmark, and face-image-quality assessment more broadly (Schlett et al.; Terhorst et al.; and systems such as OFIQ and NIST FRVT quality) produces scalar quality or per-criterion verdicts. To our knowledge none attaches a distribution-free per-criterion coverage guarantee, which is the gap we address. We build on the open-source BioGaze analyzer and the TONO synthetic dataset, and we position real multi-attribute validation (for example the DFIC dataset) as follow-up.

Note. Positioning against the most recent concurrent work is intentionally left for supervisor review; specific 2026 references should be verified before submission.

3. Method

Figure 1 summarises the pipeline: BioGaze produces per-criterion decisions and continuous metrics, a per-criterion meta-classifier maps them to a violation probability, split conformal prediction turns that probability into a prediction set with a coverage guarantee, and the 14 sets are aggregated into an accept, reject or retake decision.

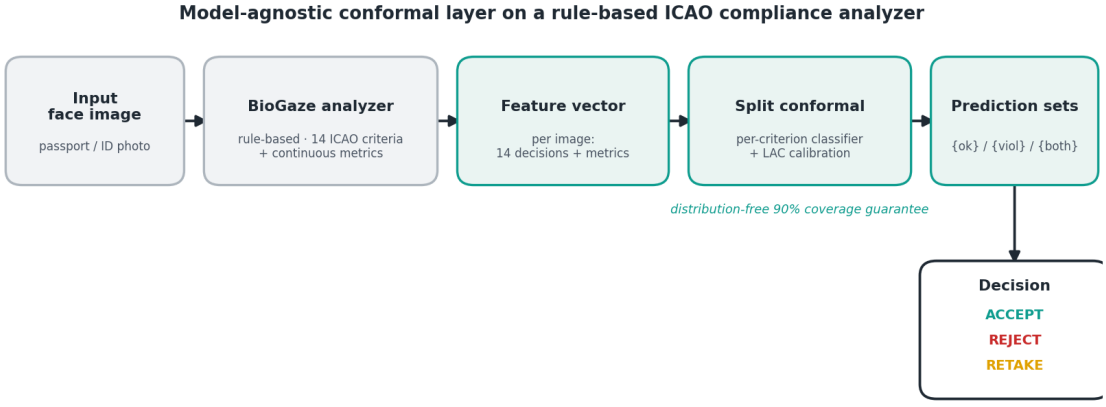


Figure 1. Model-agnostic conformal layer on a rule-based ICAO analyzer: BioGaze features, per-criterion classifier, split-conformal prediction sets, and an accept/reject/retake decision.

3.1 Features and per-criterion meta-classifier

For each image, BioGaze produces 14 binary ICAO decisions together with continuous metrics (inter-eye distance, head-pose angles, eye and mouth openness, and a glasses indicator). For each criterion we fit a class-balanced logistic regression to predict the ground-truth label, violation versus compliant, from these outputs. This is a supervised meta-classifier over all BioGaze-derived features, not a probability calibration of a single criterion's own decision, followed by conformal calibration of its scores; the analyzer itself is never retrained. Section 4.1 examines, by ablation, which features actually drive the result, since using all 14 decisions lets the classifier exploit cross-criterion structure specific to the single-violation dataset.

3.2 Subject-disjoint evaluation

The TONO subject identifier is parsed from each filename. Of 275 identities, 247 appear in more than one violation folder, so a split stratified by violation type would place the same identity in calibration and test and inflate coverage. We therefore partition identities, not images, at 60/20/20 into train, calibration and test, with zero identity overlap (verified). Every headline number is reported as a summary over multiple subject-disjoint splits (five for aggregate comparisons, ten for per-

criterion coverage). Because the ten test sets overlap (each image is a test image in about 2.4 of the ten seeds), per-criterion confidence intervals are computed on unique test images: each image's coverage indicator is averaged over the seeds in which it was tested, and a Clopper-Pearson interval is formed with the number of unique images as the sample size. This removes the seed overlap but not the residual dependence between images of the same subject (Section 6). Because a subject contributes several correlated images (median 14 to 15 per subject in every split), image-level exchangeability of calibration and test scores is an assumption of the analysis rather than a consequence of the split; we state the guarantee at the image level under that assumption and test its sensitivity to within-subject dependence in Section 4.7.

Scope of the label space. TONO's zoom folder depicts a head that is too large for the ICAO framing envelope, a head-size criterion that BioGaze does not check. These images carry no in-scope label, so they are excluded from every per-criterion coverage and trade-off computation (Sections 4.1 to 4.3 and 4.5 to 4.7) and appear only in the image-level counts of Section 4.4, where they are reported separately as out-of-scope. Every in-scope image violates exactly one criterion by construction.

3.3 Split conformal: marginal, Mondrian, and APS

We apply split conformal prediction with the LAC score s = 1 minus the predicted probability of the true class. On the calibration split we take the exact finite-sample threshold as the k-th order statistic with k = ceil((n+1)(1 minus alpha)), returning the trivial include-all set when calibration is too small to certify. A class is included in an image's set when 1 minus its predicted probability is at most the threshold. The marginal formulation uses one threshold; the label-conditional (Mondrian) formulation estimates a separate threshold per class, which enforces coverage within the minority violation class. We also evaluate a correctly randomized APS. The layer is implemented from scratch for full control and cross-checked against the MAPIE library for the marginal case (agreement within 0.003 coverage).

3.4 Triage strategies and baselines

On the same five subject-disjoint partitions we compare seven strategies: the raw BioGaze rule, the logistic classifier alone, two conventional abstention rules, marginal LAC, Mondrian, and randomized APS. The abstention rules are chosen as strong rather than weak baselines. The first is Chow's confidence-rejection rule: abstain when max(p, 1 minus p) is below a confidence level tau, otherwise decide by the majority class. The second is an asymmetric two-threshold rule that decides compliant when p is at most t_lo, decides violation when p is at least t_hi, and abstains in between, with the pair chosen to minimise a cost of ten per missed violation plus one per false flag among decided cases. Every tuned quantity is set on the calibration split only: tau is chosen so that the calibration abstention rate is closest to a fixed design target of 0.10, and (t_lo, t_hi) is chosen with calibration abstention within 0.02 of the same target. The target is a design constant equal to the nominal miscoverage alpha, the natural abstention budget of a level-alpha conformal method, and is not derived from test data; no tuned quantity in this paper uses the test split. Because Mondrian at the 0.90 level abstains on about 10% of test cases, the comparison at this target is approximately matched in abstention; Section 4.3 also reports the abstention rules at targets 0.05 and 0.15 so that the conclusion does not rest on one target. As a calibrated point baseline for the risk-coverage view we also fit isotonic regression on the calibration split and sweep its threshold. For the set-valued methods, a confident singleton violation maps to REJECT, a confident compliant to ACCEPT, and an ambiguous or empty set to RETAKE; for the abstention rules, abstention maps to RETAKE.

4. Results

4.1 The guarantee is feature-agnostic; the accuracy is a dataset artifact

Because TONO is single-violation, every per-criterion classifier can exploit the other 13 BioGaze decisions: when another criterion is flagged, this criterion is almost certainly compliant. Table 1 quantifies this. With the full feature set the logistic layer removes 66% of BioGaze's false flags; with only this criterion's own decision plus the continuous metrics the reduction falls to 26%; and with the continuous metrics alone the false-positive rate rises above the raw analyzer. Yet the conformal coverage is essentially unchanged across all three feature sets, exactly as the distribution-free theory predicts. The reading, shown in Figure 2, is that validity transfers while accuracy does not: with weak features Mondrian still reaches 0.93 violation coverage, but only by inflating its prediction sets (mean size 1.55, with 55% of sets containing both classes), which is uninformative for triage. The accuracy gains are therefore tied to the single-violation structure and should not be expected to transfer to real multi-violation data, whereas the guarantee should.

Table 1. Feature ablation (five subject-disjoint seeds). The coverage guarantee is stable across feature sets; the logistic accuracy is not, and weak-feature coverage is held only by ambiguous sets.

Feature set	Logistic FPR	False-flag redn.	Mond. viol. cov.	Mond. set size	Ambig. rate
Full (14 dec. + 7 metrics)	0.048	66%	0.902	0.98	0.04
Own decision + metrics	0.107	26%	0.906	1.05	0.09
Metrics only	0.405	-182%	0.935	1.55	0.55

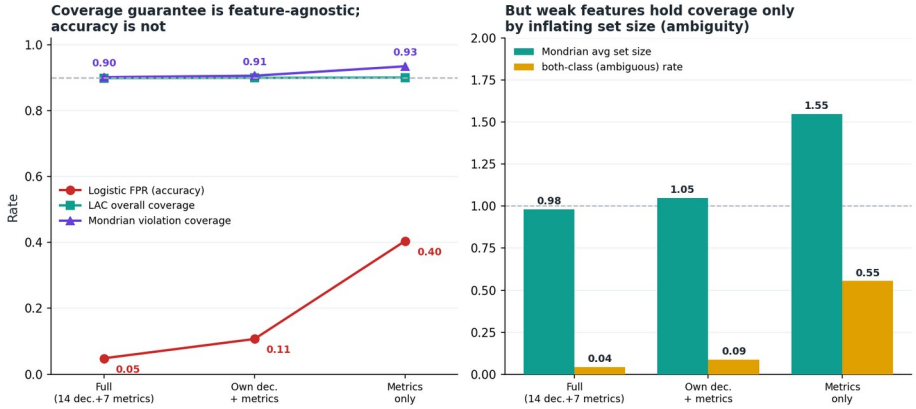


Figure 2. Left: as features are weakened, logistic accuracy (false-positive rate) collapses while LAC overall coverage and Mondrian violation coverage stay on target. Right: weak features maintain coverage only by inflating set size and ambiguity.

4.2 Marginal conformal under-covers the violation class; Mondrian restores it

Using the exact finite-sample quantile and excluding zoom images (Section 3), marginal LAC holds its overall target (about 0.90) but its violation-class point estimate is below 0.90 on 11 of the 14 criteria (mean over all 14: 0.81), and on 5 of them (6 if the small-sample Frontal pose is counted) the entire de-duplicated 95% interval lies below the 0.90 target: No posterization 0.40 (95% interval 0.34 to 0.46), No sunglasses 0.68 (0.61 to 0.74), In focus 0.73 (0.67 to 0.78), Head without covering 0.79 (0.73 to 0.84), and No pixelation 0.79 (0.74 to 0.84). Mondrian raises each of these to about 0.88 to 0.91 with intervals disjoint from the marginal method, and does no measurable harm on the criteria where marginal coverage was already adequate (Figure 3, Table 2). Excluding one degenerate small-sample criterion (Frontal pose, about 11 calibration positives), the mean Mondrian violation coverage over the 13 stable criteria is 0.90. We interpret this as restoring the intended label-conditional coverage property, that is, bringing mean violation coverage to the target, and not as a claim that every criterion exceeds 0.90 on every split (several Mondrian point estimates sit just below 0.90, which is why we report intervals and calibration counts). That Mondrian restores class-conditional coverage is expected by construction; the central empirical finding is the size of the marginal failure on a real compliance pipeline, down to 0.40 on a criterion whose overall coverage is on target.

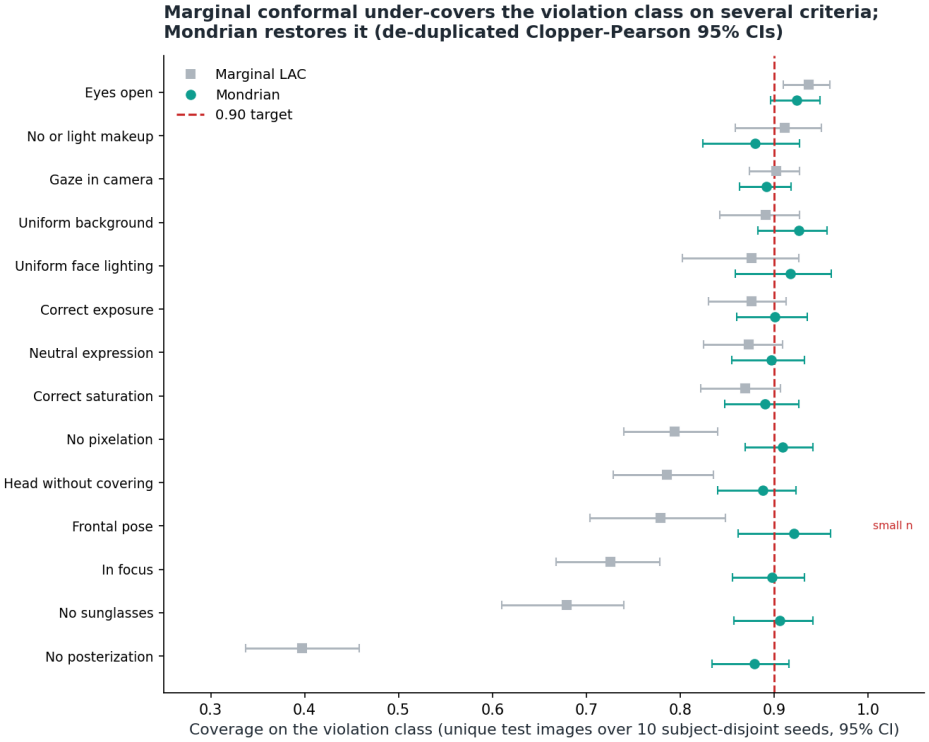


Figure 3. Per-criterion coverage on the violation class, marginal LAC (grey) versus Mondrian (teal), estimated on unique test images over ten subject-disjoint seeds with de-duplicated Clopper-Pearson 95% intervals. Frontal pose is flagged as small-sample.

Table 2. Per-criterion violation-class coverage (ten seeds; unit = unique test image, de-duplicated Clopper-Pearson 95% CI). Marginal LAC under-covers where the classifier is weak; Mondrian restores about 0.90. Calib. n is the median number of calibration positives.

ICAO criterion	LAC viol.	LAC 95% CI	Mond. viol.	Mond. 95% CI	Calib. n
No posterization	0.40	0.34-0.46	0.88	0.83-0.92	59
No sunglasses	0.68	0.61-0.74	0.91	0.86-0.94	52
In focus	0.73	0.67-0.78	0.90	0.86-0.93	60
Frontal pose*	0.78	0.70-0.85	0.92	0.86-0.96	11
Head without covering	0.79	0.73-0.84	0.89	0.84-0.92	57
No pixelation	0.79	0.74-0.84	0.91	0.87-0.94	60
Correct saturation	0.87	0.82-0.91	0.89	0.85-0.93	60
Neutral expression	0.87	0.83-0.91	0.90	0.86-0.93	60
Uniform face lighting	0.88	0.80-0.93	0.92	0.86-0.96	30
Correct exposure	0.88	0.83-0.91	0.90	0.86-0.94	60
Uniform background	0.89	0.84-0.93	0.93	0.88-0.96	51
Gaze in camera	0.90	0.87-0.93	0.89	0.86-0.92	117
No or light makeup	0.91	0.86-0.95	0.88	0.82-0.93	39
Eyes open	0.94	0.91-0.96	0.92	0.90-0.95	95

** small-sample, reported for completeness only.*

4.3 The false-flag versus detection trade-off

Table 3 compares the seven strategies on the same five subject-disjoint partitions. Three observations are essential. First, most of the false-flag reduction comes from recalibration, not conformal: a plain logistic layer already lowers the false-positive rate from 0.144 to 0.048 while keeping recall at 0.94. Second, the conformal methods do not win on accuracy. When plotted as a risk-coverage curve (Figure 4) the conformal operating points lie on or slightly below the isotonic-calibrated baseline, that is, they are matched or weakly dominated by a calibrated classifier with a swept threshold. Third, once abstention is allowed, the calibration-tuned abstention rules reach lower false-positive rates than Mondrian at a similar or lower RETAKE rate (Chow 0.027 at 9.5% RETAKE, the asymmetric rule 0.013 at 7.3%, Mondrian 0.040 at 10.3%), while recall among decided cases is comparable across all abstaining methods (0.94 to 0.96). Randomized APS is competitive (false-positive rate 0.026 at recall 0.89 over all cases); the non-randomized APS variant degenerates to trivial two-class sets on this binary problem, a property of that variant, not of APS. Because an abstaining method can improve both FPR and recall simply by routing hard cases to RETAKE, Table 3 reports recall both over all violations and among decided cases, together with the RETAKE rate and the selective error. The recall in Table 3 is measured on the common test partitions and is a distinct quantity from the per-criterion violation coverage of Table 2. Table 4 repeats the abstention rules at targets 0.05 and 0.15: the ranking does not depend on the target, and 0.10 is in fact the most favourable matched point available to Mondrian, whose abstention over the whole alpha sweep is never below about 0.10.

Table 3. Consolidated comparison of seven triage strategies on one consistent protocol (mean plus or minus SD over the same five subject-disjoint partitions). Recall (all) counts an abstained violation as missed; recall (dec.) is measured among decided cases only. Decision coverage is 1 minus the RETAKE rate; selective error is the error rate among decided cases. FPR is computed over all compliant cases, abstained ones included; Figure 5 uses decided cases only. All thresholds are tuned on calibration only (Section 3.4). No single method dominates: the asymmetric rule attains the lowest FPR and selective error, Mondrian and APS the highest recall over all cases among abstaining methods, and only the conformal methods carry a finite-sample coverage guarantee.

Method	FPR	Recall (all)	Recall (dec.)	RETAKE	Dec. cov.	Sel. error	FF redn.
Raw BioGaze	0.144 ± 0.005	0.955 ± 0.003	0.955 ± 0.003	0.0	1.00	0.137	n/a
Logistic only	0.048 ± 0.004	0.939 ± 0.012	0.939 ± 0.012	0.0	1.00	0.049	67 ± 4%
Chow rejection	0.027 ± 0.003	0.800 ± 0.010	0.954 ± 0.011	9.5	0.905	0.031	81 ± 2%
Asymmetric two-threshold	0.013 ± 0.005	0.784 ± 0.046	0.940 ± 0.015	7.3	0.927	0.017	91 ± 4%
LAC (marginal)	0.036 ± 0.004	0.807 ± 0.016	0.944 ± 0.012	7.3	0.927	0.039	75 ± 4%
Mondrian	0.040 ± 0.004	0.878 ± 0.016	0.959 ± 0.007	10.3	0.897	0.045	72 ± 4%
Randomized APS	0.026 ± 0.004	0.891 ± 0.017	0.958 ± 0.008	7.2	0.928	0.029	82 ± 3%

Tuning and set types. *Chow's tau and the asymmetric pair (t_lo, t_hi) are set on the calibration split to a fixed abstention target of 0.10; the test split is never used. Prediction-set frequencies on test are, for Mondrian, 90% singleton, 4% ambiguous and 6% empty (LAC 92.7/0.9/6.5; APS 93/7/0). Violation-class coverage on the same partitions is 0.82 (LAC), 0.90 (Mondrian) and 0.96 (APS): APS over-covers the violation class at the 0.90 level, which is why, despite its good trade-off, it is not the class-conditional recommendation.*

Table 4. Abstention rules at three calibration targets (same five partitions; Mondrian and APS do not depend on the target and are repeated for reference). RETAKE is in percent. At every target the abstention rules reach a lower FPR than Mondrian, at equal or lower RETAKE except for Chow at 0.15, which abstains more; recall among decided cases is comparable throughout, and what changes with the target is how many violations are abstained rather than detected. The asymmetric rule cannot reach a 0.15 abstention within its search grid, so its closest feasible point (0.09) is shown.

Method	Target	FPR	Recall (all)	Recall (dec.)	RETAKE	Sel. error
Chow rejection	0.05	0.035 ± 0.003	0.873	0.948	4.9	0.037
Chow rejection	0.10	0.027 ± 0.003	0.800	0.954	9.5	0.031
Chow rejection	0.15	0.021 ± 0.003	0.743	0.958	14.3	0.025
Asymmetric two-threshold	0.05	0.020 ± 0.007	0.819	0.930	5.4	0.024
Asymmetric two-threshold	0.10	0.013 ± 0.005	0.784	0.940	7.3	0.017
Asymmetric two-threshold	0.15 (0.09 reached)	0.013 ± 0.005	0.787	0.953	8.8	0.016
Mondrian (alpha 0.10)	n/a	0.040 ± 0.004	0.878	0.959	10.3	0.045
Randomized APS (alpha 0.10)	n/a	0.026 ± 0.004	0.891	0.958	7.2	0.029

Figure 5 traces each abstaining method across operating points, plotting the false-positive rate among decided cases against the abstention (RETAKE) rate. The conformal methods are plotted as points over a sweep of alpha, because their abstention rate is not monotone in alpha: it falls as alpha decreases and empty sets disappear, then rises again as sets become ambiguous. At any given abstention rate the calibration-tuned abstention rules reach a false-positive rate lower than or equal to that of either conformal method, and no method dominates once the guarantee is left out of the comparison. A conventional abstention rule therefore provides an equal or better empirical operating point. What it does not provide is a controllable class-conditional guarantee. On a binary criterion the conformal methods are themselves threshold rules: marginal LAC is, criterion by criterion, Chow's rule with tau equal to max(q, 1 minus q) for the single calibration quantile q (pooled over 14 criteria the LAC points in Figure 5 differ from the Chow curve only because LAC sets q per criterion whereas the Chow sweep uses one common tau), and Mondrian is a class-conditional two-threshold rule whose thresholds are the per-class calibration quantiles of the nonconformity score. Table 3 therefore compares two rule families, a symmetric confidence band and an asymmetric two-threshold band, each instantiated twice: with thresholds fixed as quantiles at the nominal level, or searched against an abstention or cost objective. The quantile choice is what gives Mondrian its property: its violation-class coverage is set directly by the nominal level 1 minus alpha and holds in finite samples under exchangeability, with no threshold searched against a cost objective. Figure 5 (right) shows the achieved violation coverage following the nominal target across the sweep; it sits 0.003 to 0.011 below nominal at the four levels from 0.70 to 0.90 and slightly above at tighter levels, a shortfall of the size that residual within-subject dependence or a calibration-to-test subject shift would produce, and consistent with the exchangeability caveat of Section 4.7. The abstention rules' violation coverage is instead an uncontrolled by-product of a tuned band, and marginal LAC's is systematically below target (0.82 at the 0.90 level). The contribution of Mondrian in this setting is therefore a principled recipe for choosing the two thresholds, carrying a finite-sample validity property that a tuned rule does not have, together with the demonstration that the marginal quantile fails on the minority class; it is not a win on FPR or recall.

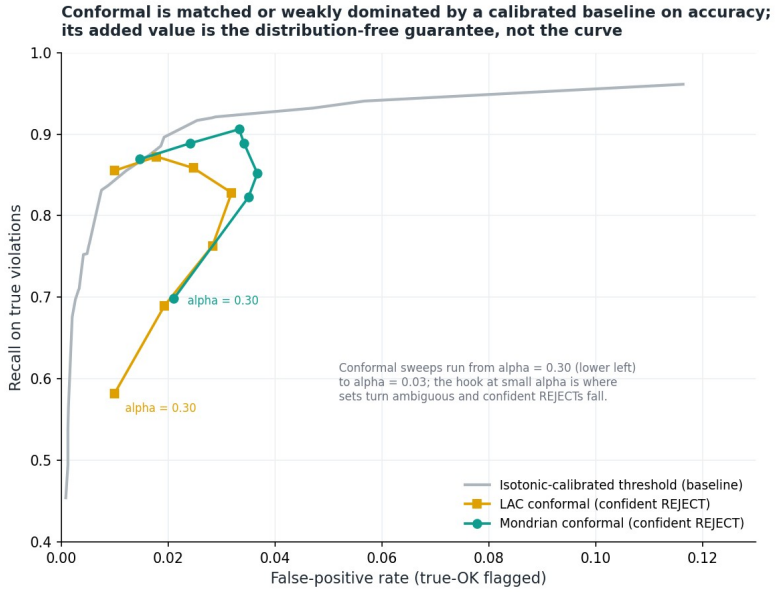


Figure 4. Risk-coverage view (partition 1). Conformal operating points lie on or slightly below the isotonic-calibrated baseline, so conformal is matched or weakly dominated by a calibrated classifier on the accuracy trade-off; its value is the guarantee, not the curve.

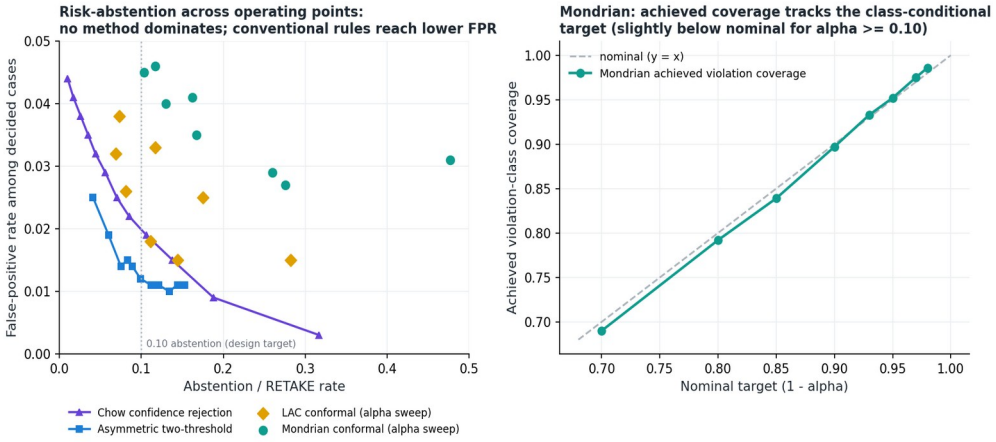


Figure 5. Left: risk-abstention operating points (five partitions pooled), false-positive rate among decided cases versus RETAKE rate; the calibration-tuned abstention rules (lines) reach an FPR lower than or equal to the conformal methods (points, alpha sweep) at any abstention rate. Right: Mondrian's achieved violation-class coverage tracks the nominal target 1 minus alpha, the property the abstention rules do not have.

4.4 Exploratory image-level triage on TONO

Aggregating the Mondrian per-criterion sets into an image decision yields 0 ACCEPT, 958 REJECT and 36 RETAKE on the 994-image test set of the first partition (Figure 6). This set contains 925 in-scope violation images and 69 out-of-scope zoom images (Section 3.2). A correct ACCEPT is structurally unreachable on the in-scope images, since each violates exactly one criterion by construction. The only available proxy for image-level specificity is the 69 zoom images, which are compliant on all 14 checked criteria; all 69 were rejected (0 of 69 accepted). Zoom images are out of distribution for the head-size metric and were excluded from training and calibration, so this proxy is weak, but it is the only one TONO offers and it did not pass. Because a naive REJECT count conflates rejecting for the right reason with rejecting for the wrong one, we decompose the outcome on the 925 in-scope violation images by whether the truly-violated criterion is the one flagged. We find 822 correct-target REJECTs (88.9%, the real violation is confidently flagged), 67 wrong-reason REJECTs (7.2%, the image is rejected but on a different criterion, so the real violation is missed even though the photo is still stopped), 36 RETAKEs (3.9%), and 0 ACCEPTs. The triage flags the correct criterion in the large majority of cases; the 7.2% wrong-reason rejections are the failure mode to report, and the empty ACCEPT class means the zero count of accepted violations carries no information about safety. We do not over-claim a general safety property: TONO is single-violation, so essentially no genuinely compliant images exist, the ACCEPT class is empty, and image-level specificity (the risk of wrongly rejecting a compliant photo) cannot be tested here. A real multi-attribute dataset with a genuine, in-distribution ACCEPT class is needed to evaluate this.

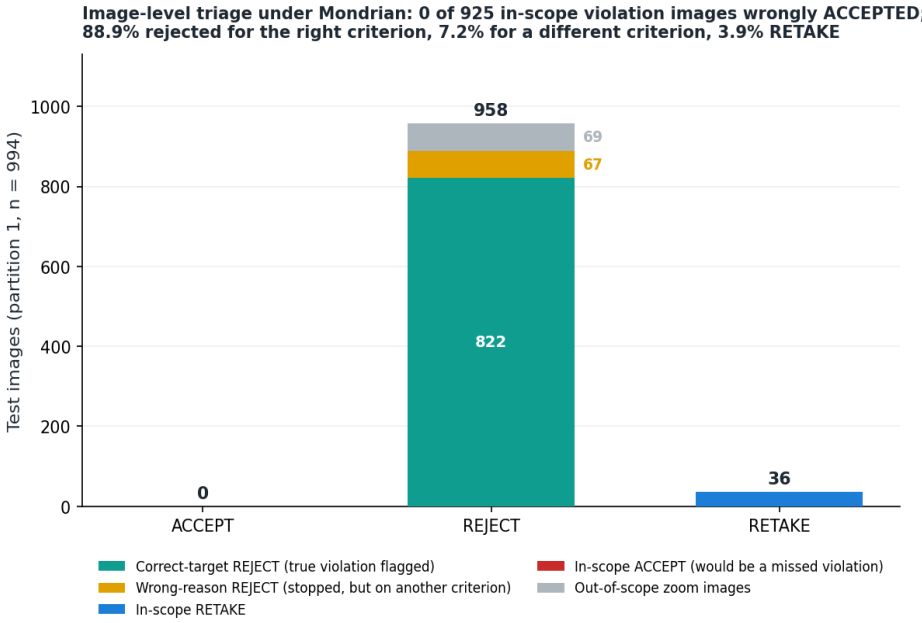


Figure 6. Image-level triage under Mondrian (partition 1), with REJECT decomposed into correct-target, wrong-reason and out-of-scope zoom images. No in-scope violation image is wrongly accepted; ACCEPT is empty because TONO contains no genuinely compliant in-scope images.

4.5 Joint coverage and the multiple-comparison problem

The per-criterion guarantees are marginal in the criteria. Requiring all 14 prediction sets to contain the truth simultaneously gives only about 0.27 empirical joint coverage, the expected multiple-comparison effect. A Bonferroni correction (per-criterion target 0.993) raises empirical joint coverage to about 0.91 for both marginal and Mondrian conformal, but the price is decisiveness. For marginal LAC about 96% of images are routed to RETAKE. For Mondrian the corrected violation-class threshold is vacuous at this calibration size: a level of 0.993 needs at least 139 calibration positives per criterion and the largest criterion has 117, so every set contains the violation class and every image is routed to RETAKE (Figure 7); its 0.91 joint coverage comes entirely from the finite compliant-class thresholds. The marginal method keeps a finite threshold only because it pools both classes (about 850 calibration images), and it attains 0.91 joint coverage by routing 96% of images to RETAKE. A Bonferroni image-level guarantee is therefore attainable but not useful at this scale; a tighter simultaneous guarantee is future work.

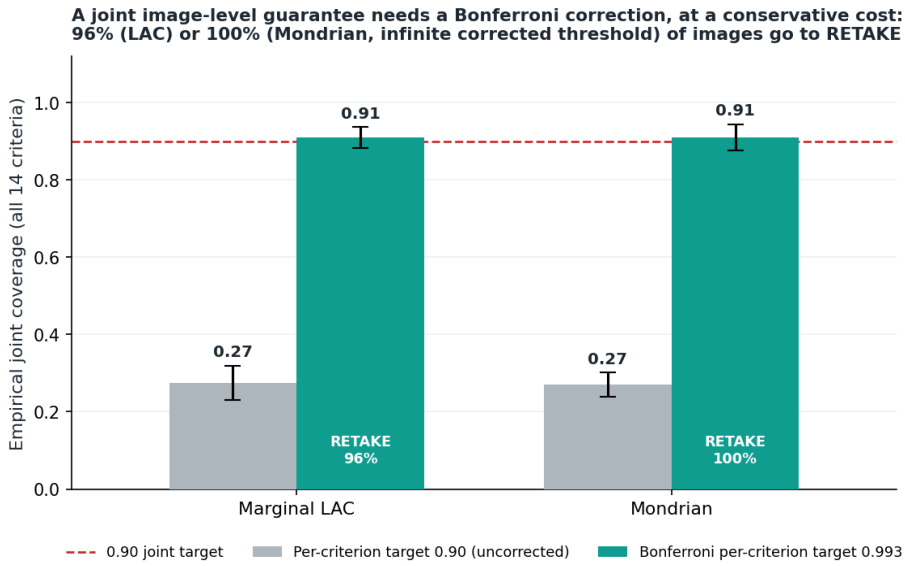


Figure 7. Naive joint coverage over 14 criteria collapses to about 0.27; a Bonferroni correction restores about 0.91 empirically, but routes 96% (LAC) or all (Mondrian, whose corrected violation threshold is infinite at this calibration size) of the images to RETAKE.

4.6 Robustness

The conclusions are stable across the checks we ran. Marginal overall coverage tracks the nominal target across levels (targets 0.95, 0.90 and 0.80 give empirical 0.951, 0.899 and 0.803), and the Mondrian-restores-violation-coverage result holds at every level of the alpha sweep in Figure 5 and on each of the five partitions individually (Mondrian violation coverage 0.88 to 0.91 versus marginal 0.80 to 0.84). Randomized APS gives the same picture across partitions and across the two independent randomization streams of our two analysis scripts (recall 0.889 and 0.891, set size about 1.07). We did not vary the split ratio.

4.7 Exchangeability and cluster-aware sensitivity

Split conformal prediction requires exchangeability of the calibration and test scores. Subject-disjoint splitting removes identity leakage, but it does not by itself establish image-level exchangeability, because a subject contributes several correlated images. We take one position throughout the paper: the guarantee is claimed at the image level under the exchangeability assumption, subject-disjoint splitting is presented as protection against identity leakage rather than as proof of exchangeability, and the sensitivity of the result to within-subject dependence is tested empirically. The structure is as follows. Every partition holds 165 training, 55 calibration and 55 test subjects; the calibration split holds 760 to 1020 images (median 14 to 15 per subject) and the test split 780 to 1080 images (median 14 to 15 per subject). The single largest subject contributes about 8% of the calibration set on average (at most 14%) and about 10% of the test set (at most 16%); no subject exceeds 16% of either split.

To test sensitivity to within-subject dependence we separate the effect of clustering from the effect of sample size. Capping each subject to at most three calibration images gives a calibration set of at most 165 images (three per calibration subject) and Mondrian violation coverage 0.94 (SD 0.02 over partitions); a random image subsample of exactly the same size, ignoring subjects, gives 0.94 (SD 0.02). The two agree, so at equal n we could not detect any clustering effect: the shift from 0.90 to 0.94 is the conservatism of the exact quantile at small n, not a subject effect. A subject-level bootstrap of the full calibration set (resampling subjects with replacement, keeping multiplicity) gives mean 0.89 with a 5th to 95th percentile range of 0.86 to 0.91, which contains the 0.90 target; the standard deviation of coverage across test subjects is about 0.12, which quantifies the heterogeneity that a subject-level guarantee would have to absorb (Figure 8). This check has limited power, because the capped and the random subsamples contain about three images per subject either way, so the contrast in clustering between the two arms is small by construction. A more direct diagnostic is the spread of Mondrian violation

coverage across the five partitions: its standard deviation is 0.010, essentially the binomial value expected under independence at about 850 violation evaluations per partition (0.010); a substantial design effect from within-subject dependence would appear as excess between-partition variance and does not. Within the power of these checks, the main result is not an artifact of within-subject dependence. A formal clustered or subject-level conformal guarantee remains future work.

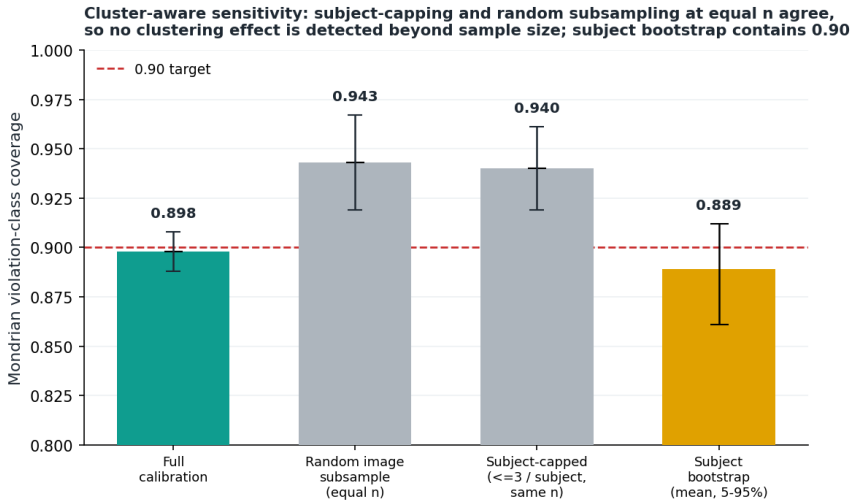


Figure 8. Cluster-aware sensitivity of Mondrian violation-class coverage. Subject-capped and random equal-n subsamples agree, so no clustering effect is detected beyond the small-n conservatism of the exact quantile; the subject bootstrap range contains the 0.90 target.

5. Auditability

Because the underlying analyzer is rule-based, each BioGaze decision can be accompanied by a region-level explanation from face-parsing segmentation and 68-point landmarks that localises the evidence the rule used (for example the lens region for sunglasses or the eyelid geometry for closed eyes). These explanations are faithful to BioGaze's own rule, not to the meta-classifier's decision: Section 4.1 shows that on TONO the meta-classifier draws substantially on cross-criterion features, which a region map does not explain. We therefore present Figure 9 as a demonstration of what can be shown to an applicant alongside a REJECT or RETAKE, not as evidence of an audited pipeline.

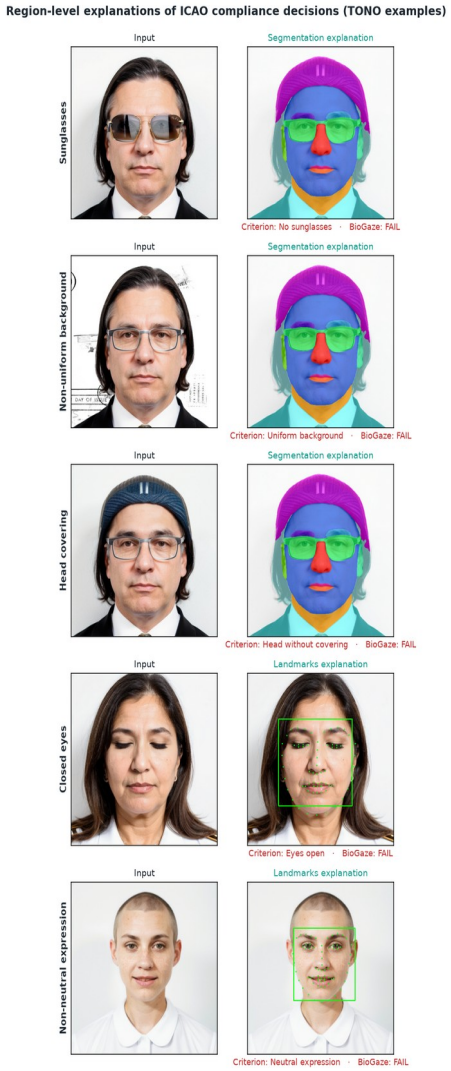


Figure 9. Region-level explanations of BioGaze's rule-based decisions on TONO examples (demonstration).

6. Limitations and discussion

- Recalibration versus conformal. Most of the false-flag reduction is achieved by recalibration alone; conformal's added value is the distribution-free guarantee and, through Mondrian, the class-conditional guarantee, not a further accuracy gain.
- Single-violation artifact. TONO's structure lets the classifier exploit cross-criterion signals, so the accuracy numbers are upper bounds tied to the dataset and are not expected to transfer to multi-violation data. The guarantee is feature-agnostic and does transfer.
- Empty ACCEPT class. TONO has no genuinely compliant in-scope images, so image-level specificity and the ACCEPT decision cannot be validated in-distribution here; the only proxy, 69 out-of-scope zoom images, was rejected in full.
- Subject-level guarantee. We provide an empirical subject-level diagnostic, not a formal clustered-conformal guarantee; the check can fail to detect a clustering effect smaller than its sampling noise.
- Validity versus usefulness. Mondrian's advantage is a validity property, not an accuracy or false-flag advantage: a calibration-tuned abstention rule reaches a lower false-positive rate at the same retake rate. Whether the formal guarantee is worth that cost is an operational decision that this study informs but does not settle.
- Confidence intervals. The intervals of Table 2 are de-duplicated over seeds (unit = unique test image, about 2.4 appearances per image), which removes the overlap between seeds; they still treat images of the same subject as independent, so they remain somewhat narrow. Under the de-duplicated intervals the five under-covered criteria stay below 0.90 and disjoint from Mondrian; under pooled intervals they would have been narrower still.
- Joint coverage. Per-criterion guarantees do not compose into an image-level guarantee without a conservative multiple-comparison correction.

- Small-sample criteria. Coverage estimates for rare criteria (for example Frontal pose) are unstable and are flagged rather than relied upon.

These limitations motivate a follow-up study on a real, multi-attribute dataset (for example DFIC) with genuinely compliant images: re-running the identical pipeline under a subject-disjoint split would allow the full accept/reject/retake confusion to be evaluated, a cost-calibrated retake region to be tuned against the asymmetric miss and retake costs, and a corrected simultaneous guarantee to be reported.

7. Conclusion

We showed that a rule-based ICAO compliance analyzer can be wrapped with a per-criterion conformal layer that supplies a distribution-free coverage guarantee under image-level exchangeability, evaluated under a strictly subject-disjoint protocol. The central empirical finding is the size of the marginal method's failure on the minority violation class, down to 0.40 on a criterion whose overall coverage is on target, and its repair by label-conditional Mondrian conformal, with disjoint intervals on the five criteria where the marginal method fails. A feature ablation separates the transferable guarantee from single-violation accuracy, and a seven-method comparison with calibration-only tuning shows that recalibration provides most of the accuracy, that a conventional abstention rule can provide an equal or better empirical operating point, and that Mondrian's distinct contribution is the formal class-conditional coverage property. We characterise the multiple-comparison problem of a 14-criterion decision and confine every claim to what the dataset can support. Future work will validate on real multi-violation data and pursue a cost-calibrated, simultaneously-valid triage.

References

[1] International Civil Aviation Organization. Doc 9303, Machine Readable Travel Documents, 8th ed. ICAO, 2021.

[2] ISO/IEC 39794-5:2019, Information technology, Extensible biometric data interchange formats, Part 5: Face image data. ISO, 2019.

[3] M. Ferrara, A. Franco, D. Maio and D. Maltoni. Face image conformance to ISO/ICAO standards in machine readable travel documents. IEEE Transactions on Information Forensics and Security, 7(4):1204-1213, 2012.

[4] G. Borghi et al. TONO: a synthetic dataset for ICAO compliance of face images. ECCV 2024 Workshops. doi:10.1007/978-3-031-91907-7_1.

[5] BioGaze: an open-source ICAO/ISO face-image compliance analyzer. FG 2025 (demo).

[6] V. Vovk, A. Gammerman and G. Shafer. Algorithmic Learning in a Random World. Springer, 2005 (2nd ed. 2022).

[7] A. N. Angelopoulos and S. Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511, 2021.

[8] M. Sadinle, J. Lei and L. Wasserman. Least ambiguous set-valued classifiers with bounded error levels. Journal of the American Statistical Association, 114(525):223-234, 2019.

[9] Y. Romano, M. Sesia and E. Candes. Classification with valid and adaptive coverage. NeurIPS, 2020.

[10] T. Ding, A. N. Angelopoulos, S. Bates, M. I. Jordan and R. J. Tibshirani. Class-conditional conformal prediction with many classes. NeurIPS, 2023 (arXiv:2306.09335).

[11] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei and T. Schuster. Conformal risk control. arXiv:2208.02814, 2022.

[12] S. Bates, A. Angelopoulos, L. Lei, J. Malik and M. I. Jordan. Distribution-free, risk-controlling prediction sets. Journal of the ACM, 68(6), 2021.

[13] C. K. Chow. On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory, 16(1):41-46, 1970.

[14] R. El-Yaniv and Y. Wiener. On the foundations of noise-free selective classification. Journal of Machine Learning Research, 11:1605-1641, 2010.

[15] Y. Geifman and R. El-Yaniv. SelectiveNet: a deep neural network with an integrated reject option. ICML, 2019.

[16] T. Schlett, C. Rathgeb, O. Henniger, J. Galbally, J. Fierrez and C. Busch. Face image quality assessment: a literature survey. ACM Computing Surveys, 54(10s):1-49, 2022.

[17] P. Terhorst et al. Pixel-level face image quality assessment for explainable face recognition. arXiv:2110.11001, 2021.

[18] V. Taquet, V. Blot, T. Morzadec, L. Lacombe and N. Brunel. MAPIE: an open-source library for distribution-free uncertainty quantification. arXiv:2207.12274, 2022.